

CORE 1B

Motion in 2D - Projectile Motion

Reconstruction self-tutor: predict and attempt before reading the repair or model response.

Rule

For every concept, complete PREDICT and ATTEMPT first. The reconstruction and boundary tests come after your commitment.

1. R1 - Independent components / shared clock

PREDICT

At one instant, may the x equation use $t=2$ s while the y equation uses a different time?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

No. Separate equations, one object, one clock.

BOUNDARY TEST

- If $x(2\text{ s})$ is known and $y(3\text{ s})$ is known, can those values form one position state?
- What must be common before x and y components are recombined?

Your boundary-test response

2. R2 - Constant acceleration per component

PREDICT

If $\mathbf{a}=(10,4) \text{ m/s}^2$, should both component equations use $|a|=\sqrt{116}$?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

No. Use $a_x=10$ only on x and $a_y=4$ only on y.

BOUNDARY TEST

- If a_x is constant but a_y changes, are constant-acceleration formulas valid on both axes?
- If an x-event occurs at $t=3$ s, what t belongs in the y equation for that same event?

Your boundary-test response

3. R3 - Projectile model gate

PREDICT

At the apex of an ideal projectile, which of v_x , v_y , and a_y are zero?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

Only v_y is zero; v_x remains and $a_y = -g$.

BOUNDARY TEST

- For a projectile launched from a roof to lower ground, is the same-height time-of-flight shortcut valid?
- If horizontal launch speed doubles from the same cliff, does the ideal fall time change?

Your boundary-test response

4. E1 - Velocity direction from position laws

PREDICT

For $x(t)$, $y(t)$, should direction of motion come from y/x or from $(dy/dt)/(dx/dt)$?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

Use derivatives: instantaneous velocity is tangent to the path.

BOUNDARY TEST

- Why is the position-ray direction generally different from the velocity direction?
- What must be shared when v_x and v_y are evaluated?

Your boundary-test response

5. E2 - Energy-loss bounce

PREDICT

If 50% of kinetic energy remains after impact, does the speed become 50% of its old value?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

No. $v_{\text{after}}^2 = 0.5 v_{\text{before}}^2$, so $v_{\text{after}} = v_{\text{before}} / \sqrt{2}$.

BOUNDARY TEST

- After the collision, which component controls the next rise height?
- What event separates the old projectile state from the new one?

Your boundary-test response

6. E3 - Piecewise gravity

PREDICT

Gravity changes only after the apex. Should the new gravity be used before the apex?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

No. Preserve the apex state and apply the new acceleration only to stage 2.

BOUNDARY TEST

- Which state quantities are continuous at the apex boundary?
- If $g_2 = g_1$, what result should the two-stage model reduce to?

Your boundary-test response

7. E4 - Trajectory through a point

PREDICT

If P satisfies $y(x)$, does that alone tell whether P is before or after the apex?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

No. Use the sign of v_y or local slope as a separate branch test.

BOUNDARY TEST

- Can the same height occur on both the rising and falling branches?
- What equation checks point membership?

Your boundary-test response

8. E5 - Linear drag

PREDICT

If $F_x = -c v_x$, may we still use $x = v_{x0} t$ exactly?

Your prediction + reason

ATTEMPT

Write the governing model/representation and one executable relation before looking ahead.

Your reconstruction attempt

DIAGNOSE -> REPAIR

Model response

No. v_x decays exponentially, so displacement is the integral of that decay.

BOUNDARY TEST

- What happens to the drag expression as c tends to zero?
- Why is constant-acceleration kinematics invalid on the dragged axis?

Your boundary-test response